Internal consistency, not plausibility, decides whether AI agents relay database errors

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Abstract

Agents that query scientific databases inherit their errors unless they override them. We tested when they do in a live deployment of Biomni, a ten-model benchmark and 79 protein sequences that UniProt has since superseded. Whether an agent overrode a wrong value depended on whether another field of the record contradicted it. Records that agreed with themselves were relayed whatever the error's size: superseded sequences in 90–100% of runs, with or without the protein's identity. Given a contradiction, override grew with its size and differed by model. Asked to list concerns, agents flagged most errors yet relayed many; unprompted, they rarely questioned plausible values. Instructions helped only where the record contradicted the value. Withholding fields that failed a derivation check, or silently correcting records from a current copy, left nothing to relay (0 of 709 and 0 of 702 runs). Conflicts should be resolved before the agent reads the record.

Language-model agents are moving from answering questions to doing research. Systems such as Coscientist [1], ChemCrow [2] and Biomni [3] plan analyses, query scientific databases and compute on what they retrieve; some verify their own output against curated databases [4], which presumes that the databases are right. Their answers can be no better than the records they read, and those records change: misannotation is widespread in major protein databases [5–8], and UniProt [9] archives every earlier version of every entry [10]; of the 300 reviewed human proteins in our benchmark, 167 have at least one superseded sequence. An agent that relays a database value inherits whatever is wrong with it unless it notices the problem and acts on it.

How language models treat wrong information in their context has been studied mainly as a conflict between the context and the model's own knowledge [11, 12]. Models often adopt incorrect retrieved content, less often the further it departs from the truth [13, 14], and accept coherent counter-evidence against their memory [15], and prompting can shift the balance toward the context [16]; they are distracted by irrelevant context [17, 18], steered by poisoned documents [19] and sycophantic toward stated beliefs [20, 21]. Tool use is well developed [22–25] and benchmarked [26, 27], adversarial tool returns have been studied as prompt injection [28–30], and concurrent work shows that tool-using agents adopt corrupted returns and may question a return while passing it on [31]. These studies corrupt passages of text or single returned values. A scientific database record is different: a value sits among fields that can confirm or contradict it, and the record often contains what is needed to derive it, as a sequence fixes a protein's length and mass and a formula fixes a compound's weight.

We asked what decides whether an agent overrides a wrong value in such a record. We ran a deployed biomedical agent against live database APIs with responses altered in transit; built a benchmark of protein and small-molecule questions for ten models from five developers; varied whether the record contradicted the wrong value, and by how much; presented 79 sequences that

UniProt has since superseded; separated the record's role from the model's knowledge, its reasoning and its instructions; and compared ways of correcting the record. Predictions were fixed in dated repository commits before each experiment, and the deployment hypotheses were registered on OSF; we report those that failed.

The deciding variable was internal consistency. A record that agreed with itself carried an error of any size, including real superseded entries, with or without the protein's identity. A contradiction within the record made override possible, to a degree that depended on its size, on the model and on what the model knew about the entity. Instructions and output formats changed relay only where a contradiction existed; removing the wrong value before the agent read the record was the one remedy that worked everywhere.

Results

Agents relay database errors in deployment and in a benchmark

To test whether the problem arises outside a benchmark, we ran Biomni [3], an open-source biomedical agent, against the live UniProt and PubChem APIs and altered one field of selected responses in transit with an intercepting proxy (Methods). Each of 300 tasks ran without corruption, with an impossible value and with a plausible one (a number within 25% of the true value, a donor count changed by one, or another real entity's identifier), for two models, under hypotheses registered in advance (OSF jmwab). The first run exposed the study's name to the agent through a directory path, and models can recognise evaluations from such cues [32]; we report a rerun in an isolated environment (Methods; Supplementary Note 1).

The corrupted value reached the agent in 559 and 583 of 600 corrupted runs (Fig. 1b). Impossible values were almost always caught (relayed in 4 of 293 delivered runs with DeepSeek V4.1 Flash and 4 of 283 with Gemini 3.8 Flash); plausible values were relayed in 17% (50/290) and 21% (57/276). Contrary to our registered prediction, fields with an exact derivation route were relayed more often, not less: protein length and mass in 36–62% of plausible runs, organism and gene symbol in 0–8% (odds ratios 17.2 and 6.9; Fig. 1c). Organism and gene symbol are repeated elsewhere in a UniProt record, so a corrupted copy contradicts the record; a scaled length contradicts only the sequence, which the agent must count. Gemini 3.8 Flash computed length or mass from the sequence in about half its runs and relayed half as often when it did (length 23% against 50%, mass 46% against 79%); DeepSeek never computed and relayed 37–52%.

To separate detection from relay, we built a benchmark of 300 human proteins from UniProt [9] and 300 compounds from PubChem [33], with lookup and threshold questions over ten templates, and ran ten models from five developers through a lookup tool (Methods). Clean accuracy was 98–100% for every model. In corrupted runs one field of the target record held an impossible or a plausible value; plausible numeric corruptions scaled paired fields together, so the record stayed consistent, whereas impossible values changed one field and contradicted its partner. Asked, as in our registered prompt, to list anything that looked wrong, agents flagged impossible values in 75–100% of runs yet relayed them in 32–93% of protein and 38–91% of chemistry runs, and relayed plausible values in 74–100% and 93–100% (Extended Data Fig. 1). The flags were specific: in impossible-value protein runs, 81% cited a contradiction between fields of the record and 59% the agent's memory of the protein (Methods). Without the request for concerns, agents still voiced doubt about most impossible values (36–84% of runs, scored by a classifier validated against human labels at $\kappa = 0.84$) but rarely about plausible ones (0–29%, against 43–97% when asked), and relay rose in six of twelve cells, by up to 43 percentage points, and fell in one (Methods). Detection of plausible errors in the benchmark is therefore largely elicited by the prompt, and a plain prompt

relays more.

A consistent record carries an error of any size

If agents act on contradictions, an error that contradicts nothing should pass. We edited protein sequences, or substituted compound structures, and recomputed every derived field, so that the record was wrong but internally consistent and passed every derivation check (Methods). With the full record in view, Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash relayed these coherent errors in 94–98% of runs that shortened a protein, 70–96% of runs that lengthened one and 93–98% of chemistry runs, against 9% for Opus 5 when the full record contradicted the value. From the summary record, size did not protect: a three-residue sequence presented as a known human protein, with matching length and mass, was relayed in 94–100% of runs (12–42% with the full sequence and annotation in view), and a methane structure substituted for a drug in 45–100%.

Real records arrive in this form. For each benchmark protein whose sequence UniProt has revised with a change in length, we presented the most recent superseded sequence with its own length and mass (79 proteins; Methods). The revisions ranged from 1 to more than 50 residues; 13 trace to submissions that the current entry flags as erroneous, mostly with a wrong initiation codon, 16 of the old sequences remain current isoforms of the entry, and 48 were retired without a recorded reason. Without a second source, Opus 5, Argon and DeepSeek relayed the superseded value in 90–100% of runs with the full record and in every run from the summary record (Fig. 2a), across three samples (87–100%) and whatever the size of the revision (47 of 48 full-record runs for changes of more than 50 residues, 78 of 81 for changes of 1–5; Fig. 2b). The documented errors were relayed as often as the rest (86% of full-record and 100% of summary-record runs). Removing the protein's identity changed nothing: with the accession, name and gene replaced by an opaque handle and the record reduced to organism, length, mass and sequence, the superseded values were relayed in 97–100% of runs.

A contradiction makes override possible; its size, the model and the model's knowledge decide it

Given a contradiction, agents did override. With the full record, organism errors, which contradict the entry's taxonomy, fell from 100% to 0% relay for all three models tested, and Opus 5's relay of impossible and plausible length and mass errors fell from 33% to 1% and from 75% to 9%. To measure how large a contradiction must be, we scaled one field and its paired field by factors from $\times 0.005$ to $\times 200$, so that only the sequence or formula disagreed (Methods). Every model overrode large contradictions, but the change at which half of runs overrode differed (Extended Data Fig. 2): within 20% of the true value for Opus 5, Argon and Gemini 3.8 Flash on proteins, about twofold for Sonnet 5 and DeepSeek ($\times 0.43$–0.53 when decreased, $\times 1.80$–2.31 when increased), and up to about fivefold for Sonnet 5 on molecular weight. The boundaries are specific to the models and items tested: they did not transfer to 60 new compounds in a registered follow-up (5 of 10 prediction intervals met).

The benchmark agents had no code tool, so a contradiction with a sequence was visible only to an agent that counted residues as it read, a task at which language models are unreliable [34], or that knew the protein. Three registered controls asked what else override drew on (Fig. 3). First, identity: on 25 length questions whose full record contradicted a plausible wrong length, Opus 5 relayed the value in 4% of runs with the full record, 28% when the record kept the accession, name and gene but dropped the annotation, and 68% when identity was removed as well, whereas Argon relayed it in 0–4% in all three formats and DeepSeek in 84–92%. Opus 5's override drew mainly

on what it knew about the protein and secondarily on feature annotations whose coordinates can contradict a wrong length; Argon's came from the sequence. Second, reasoning: with thinking disabled, Opus 5's relay of plausible values that the full record contradicted (first 50 benchmark tasks) rose from 8% to 46% ($p = 2 \times 10^{-6}$), whereas values that contradicted their paired field directly were not relayed more often (Sonnet 5, 70% and 63% without thinking against 76% and 73% with it) and consistent values were relayed at the same high rate. Third, memory: asked without tools for the length of the 79 historical proteins, the three models gave today's value for 61–67% and the superseded value for 8–9%, yet proteins whose remembered length was correct were still relayed in 91–100% of runs without a second source. A model's knowledge decides whether a visible contradiction is acted on, but does not by itself overturn a consistent record.

A derivation route lets agents see an error; agreement among fields decides whether they act

We registered a test with three chemistry fields chosen by how exactly they follow from the record's formula and structure: monoisotopic mass (exact), topological polar surface area (approximate) and PubChem complexity (no route; Methods). The route changed detection: agents raised a concern about a corrupted monoisotopic mass in 36–60 of 60 runs per model, against 0–6 of 60 for uncorrupted records, about TPSA errors in 0–47 of 60, and about complexity errors no more often than for clean records (Fig. 4a). It barely changed relay (96%, 99.6% and 100% across four models), and the registered prediction that monoisotopic-mass relay would follow each model's molecular-weight boundary failed (90–100% observed, 27–59% predicted). In flagged runs agents computed the correct mass from the formula, stated it, and answered from the record. A value that looked like a placeholder was treated differently: a monoisotopic mass of 0 Da was relayed in 0–3% of runs, with agents substituting the computed value.

The corrupted monoisotopic mass had agreed with two other fields of the record, whereas the molecular-weight boundaries were measured with one. A follow-up varied agreement on the same 60 compounds, three samples per run (Fig. 4b,c). Each derived field that agreed with a wrong molecular weight raised its relay, by 2–7 points for the first (heavy-atom count) and 3–28 points for the second, and corrupting the monoisotopic mass alone, so that it disagreed with the record's own molecular weight, lowered relay by 14–33 points ($p < 10^{-6}$ in the first sample for both effects). Agreement is not repetition (Fig. 5d; Methods). A verbatim copy of the wrong weight under a familiar field name (average mass) left relay where it was (64% against 63% pooled over four models), whereas the same copy left at the correct value removed most relay (9% against 63%). Under an unfamiliar field name the agreeing copy lowered relay (68% to 52%), the opposite of our registered prediction; the familiar name removed most of that drop.

What removes relay

Instructions removed relay only where the record contradicted the value (Fig. 5a–c). On the chemistry cell in which three fields agreed with a wrong weight but the formula contradicted it, dropping the registered instruction to use the provided tools lowered relay from 80–91% to 34–51% for three of four models ($p \leq 4 \times 10^{-9}$), and a sentence stating that the database contains errors and that the formula is authoritative lowered it to 0–1% for all four. On coherent protein errors, which nothing in the record contradicts, no instruction changed relay (92–100%). An output format that asked for the database value and the agent's own estimate separately did reach coherent errors for two models (relay 98% to 52% for Opus 5, 96% to 0% for Argon; DeepSeek 94% to 90%), presumably by licensing the memory that the probe shows these models hold, but it gave no reliable

audit trail: the stated database value matched the record in only 0–28% of runs on the protein cell. Memory decided the matter once a conflict was visible: with a second source shown, proteins whose remembered length was the superseded one were relayed far more often than those whose remembered length was today's (Opus 5, 13 of 21 against 2 of 144; DeepSeek, 17 of 18 against 38 of 150; $p \leq 10^{-8}$; Fig. 5e).

A harness can act on the record before the agent sees it (Methods). Checking derivable fields and annotating mismatches removed constructed single-field errors almost entirely (0–1% relay for proteins; 12–24% in chemistry, concentrated in compound names, which have no derivation route), but coherent and superseded errors pass such checks by construction. For those we compared each record with a second copy: AlphaFold DB [35, 36], which takes its sequences from current UniProt and so tests a stale record against a current one rather than providing independent evidence, and NCI CACTUS for compounds, which agreed with PubChem's formula for 270 of the 283 compounds it resolved. Where the second copy covered the entity it caught most coherent errors, and on the superseded records it cut relay to 9%, 1% and 33% for Opus 5, Argon and DeepSeek over three samples, with residual relay higher where the superseded sequence was longer than today's (23% against 11%). How the correction was presented mattered as much as whether it was made (Fig. 6). Writing the corrected value into the record while listing the original in a note left 3–11% relay; labelling the original explicitly as the original database value made Argon choose it in 41% of runs; correcting the record silently left nothing to relay (0 of 702 runs), with accuracy at 99% (234 of 237 runs per model, the misses all on the one protein the second copy did not cover). Without any second source, a harness that withheld every derivable field failing its check, with no note, left 0 of 709 corrupted numeric fields relayed, with accuracy of 93–100% on proteins (DeepSeek 79–84% if its unparsed runs are counted as wrong) and 83–95% in chemistry, where the remaining errors were compound names, which are not withheld (Fig. 5f).

Predictions that failed

Several registered predictions failed, and Methods reports each with its test: fields with an exact derivation route were relayed more, not less, in deployment; monoisotopic-mass relay did not follow the molecular-weight boundaries, and the boundaries did not transfer to new compounds; the tool instruction mattered more than predicted; a repeated value did not act as agreement; Opus 5's override of contradicted lengths depended on identity; and the keyword detector registered for unprompted flags failed validation against human labels ($\kappa = 0.39$) and was replaced by a validated classifier.

Discussion

In every setting we tested, from a deployed agent on live APIs to ten models on a benchmark to records that UniProt itself has superseded, the variable that decided whether a wrong value reached the answer was whether the record contradicted it. A record that agreed with itself carried an error of any size from the summary record, a three-residue protein as readily as a revision of a few residues, with or without the entity's identity and with or without a request to list concerns. A contradiction made override possible. Whether it happened then depended on the size of the contradiction, on the model and on what the model knew: Argon acted on a sequence alone, Opus 5 mainly on its knowledge of the protein, and DeepSeek rarely. Plausibility mattered at the extremes, where a mass of 0 Da read as a placeholder, but within the range of values that look like data it did not decide.

Two findings qualify the picture of agents that notice errors and relay them anyway. First, much of the noticing of plausible errors was elicited: asked to list concerns, agents flagged them; unprompted, they rarely did, and relayed more. Benchmarks that solicit concerns therefore overstate how often agents notice errors and understate how often they relay them. Second, agreement is not repetition. A derived field that agreed with a wrong value strengthened it, a field that disagreed removed it, and a verbatim copy under a familiar name did neither. The agents responded to contradiction, not to the number of times a value appeared.

These results extend work on knowledge conflicts, in which models adopt incorrect context even against their memory [12, 13, 15], and concurrent work showing that tool-using agents question a corrupted return while passing it on [31]. A structured record adds what text lacks: fields that can contradict one another. That structure is what the agents used, and it is what real errors lack, because a database revision replaces a sequence together with everything derived from it.

Three implications follow for systems that put agents on scientific databases. First, instructions and output formats help only where a contradiction exists. A sentence naming the formula as authoritative removed relay of contradicted values, and a format that separates the record's value from the agent's estimate let a correct memory act on consistent errors, but neither reaches an error that is consistent with its record and absent from, or wrong in, the model's memory. Second, the reliable remedies act on the record before the agent reads it. Withholding fields that fail a derivation check, or correcting silently from a current copy, left nothing to relay without lowering accuracy, whereas showing the agent both values reopened the decision and let memory or a label settle it. The design principle is to resolve conflicts before the model sees them and to keep the audit trail for the people responsible for the analysis. Third, a second copy is only as good as its currency and coverage: AlphaFold DB caught superseded UniProt sequences because it mirrors the current release, a lagging mirror would carry the same error, and entities outside its coverage remain blind spots.

The study has limits. Most conditions ran one sample per task, although the cells behind the main claims ran three. We studied two domains, two primary databases and questions with a single checkable answer. Most errors were constructed; the 79 superseded records are real, but only 13 are documented errors. The deployment used one agent framework and two models, and Anthropic models could not run in it. Behaviour is tied to the 2026 model versions tested, which were requested by alias, and reasoning settings, which we varied for two models, moderate override where it depends on derivation. Several experiments were designed after earlier results and registered before they ran.

The agents behaved less like credulous readers than like careful clerks: they recorded what the database said and corrected it when the database contradicted itself. For an agent whose answer feeds an analysis, that means database errors become analysis errors unless the record is corrected before the agent reads it.

Methods

Terms

A corrupted value is *relayed* (in the code and data files, *propagated*) when the agent's answer matches the answer the corrupted record implies, and *overridden* when the answer is the true one instead; a run is *flagged* when the agent lists a concern, and a corrupted value is *delivered* when it reached the agent. A record is *consistent* (in the constructed experiments, *coherent*) when none of its fields contradicts another, and *contradicted* when its sequence, formula or another field disagrees with the value. *Superseded* sequences are earlier UniProt versions of an entry; *documented*

errors are those traced to a submission that the current entry flags as erroneous. Harness modes: *annotate* lists each mismatch in a verification note; *replace* also writes in the corrected value, with the original listed under ambiguous field names, labelled as the original database value, or not shown at all (*silent*); *withhold* removes a mismatched derivable field without a note.

Benchmark

Entities and tasks. The benchmark draws on 300 reviewed (Swiss-Prot) human proteins, sampled with a fixed seed from a pool of 3,000 retrieved through the UniProt REST API, and 300 single-component PubChem compounds with a common name, a molecular formula and a molecular weight of 100–900 Da, sampled with a fixed seed from CIDs 1–30,000. It has 200 protein tasks over five templates (gene symbol, 50; organism, 50; mass comparison, 50; length threshold, 25; length ratio, 25) and 100 chemistry tasks over five (name, 25; hydrogen-bond donors, 25; heavy-atom threshold, 25; weight threshold, 12; weight comparison, 13). Each task has a single checkable answer and is constructed so that the answer implied by a corrupted record differs from the true answer; tasks were kept only if the true answer lay at least 5% from the decision boundary.

Tools and records. Protein agents had two tools: `get_protein`, which looks up a record by UniProt accession, and `find_protein_by_gene`, which looks up the human protein encoded by an HGNC gene symbol. Chemistry agents had `get_compound` (by PubChem CID) and `find_compound_by_name`. The summary record lists, for proteins, accession, protein name, gene symbol, organism, length, mass and subcellular location, and for compounds, CID, name, molecular weight, heavy-atom count and number of hydrogen-bond donors. The full record is the complete UniProt entry, including sequence and feature annotations, or, for compounds, the summary fields plus IUPAC name, molecular formula and SMILES. The full-record arm ran 100 of the protein tasks, spanning all five templates, with Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash, and 100 chemistry tasks with Gemini 4 Argon and DeepSeek V4.1 Flash.

Conditions. In the clean condition all records are returned unchanged. In the impossible condition the target field of the target record holds a value no real entity can have: a length of 3 or 60,000 residues; a mass of 45 Da, a negative mass or 9×10^9 Da; the organism *Escherichia coli* K-12 for a protein located in a eukaryote-only compartment; or a fabricated gene symbol. For compounds, impossible values are a molecular weight of 9×10^9 Da, 2 Da or the negative of the true weight; 0 or 50,000 heavy atoms; -3 or 500 hydrogen-bond donors; and a fabricated name. Impossible numeric values change a single field and therefore also contradict the record's paired field. In the plausible condition a protein's length and mass are scaled together by 0.75–0.8 or 1.2–1.25, so the summary record stays internally consistent, and compound weights and heavy-atom counts are scaled in the same way; identifiers are replaced by another real entity's value (another protein's gene symbol, the organism *Mus musculus*, or another benchmark compound's name), and a compound's hydrogen-bond donor count is changed by one. Directions are balanced. Only the target record is altered; every other record in a run is returned unchanged. In the verify arm, one sentence was appended to the system prompt.

Prompt. The system prompt read: "You are a bioinformatics assistant. Use the provided tools to answer. End your reply with exactly one line of JSON: {`"answer": ..., "concerns": [...]`} where concerns lists anything about the data that looked wrong or surprising (empty list if nothing)." For chemistry, "bioinformatics" was replaced by "chemistry". The verify sentence read: "Before answering, sanity-check every tool result against your biological knowledge and report inconsistencies in concerns."

Models and inference

We evaluated ten models: Claude Haiku 4.5, Sonnet 4.6, Sonnet 5 and Opus 5 (Anthropic API); Gemini 3.8 Flash and Gemini 4 Argon (Google Vertex AI); GPT-6 Luna and GPT-6.1 Sol (OpenAI API); and DeepSeek V4.1 Flash and Kimi K3 (Together AI). In each run the model received the system prompt, the task question and the tool schemas, could call the tools in any order and any number of times, and ended with the JSON line. We used each provider's default sampling settings and allowed 8,000 output tokens per request. Anthropic models ran with adaptive thinking (disabled only in the reasoning ablation); for other providers no reasoning parameter was sent, so each ran at its provider's default reasoning setting. Requests named the models by the aliases below; dated snapshot identifiers were not recorded. The request settings recorded in the traces are listed for every model in Supplementary Table 16. Unless stated otherwise, each task ran once per condition. API identifiers: `claude-haiku-4-5`, `claude-sonnet-4-6`, `claude-sonnet-5`, `claude-opus-5`; `gemini-3.8-flash`, `gemini-4-argon`; `gpt-6-luna`, `gpt-6.1-sol`; `deepseek-ai/DeepSeek-V4.1-Flash-1017`, `moonshotai/Kimi-K3`. All runs took place between September and November 2026.

Outcomes

The answer and concerns were parsed from the final JSON line. A run is correct if its answer matches the true answer and propagated if it matches the answer implied by the corrupted record; because every task separates the two, a parsed answer of a corrupted run is at most one of these. Yes/no answers match on the first yes or no in the answer, numbers within 5%, and text case-insensitively, either exactly or as a contiguous run of words within the answer. A run is flagged if its concerns list is non-empty. Runs without a parseable answer were excluded and counted, and runs that failed at the API were retried. Clean accuracy was 98–100% for every benchmark model and 100% in the full-record arm. Counting unparsed runs against the agent changed no reported rate by more than three points except in three cells: Opus 5 with the errors-noted prompt on the protein cell (0 of 38 relayed, 12 unparsed), Gemini 3.8 Flash with the two-fields format (6 of 163, 17 unparsed) and DeepSeek in the withholding arm on proteins (16 and 13 unparsed; Supplementary Table 20).

Statistics

Proportions are reported with Wilson [37] 95% intervals and, where tasks repeat across models, task-cluster bootstrap intervals [38]. Benchmark effects were estimated by logistic regression with task-clustered standard errors. Registered experiments used the tests named in their registrations: Fisher's exact test, McNemar's test, the Wilcoxon signed-rank test or logistic regression, one-sided where the prediction was directional and Holm-corrected where registered.

Run cache and integrity

Runs were cached by model, condition, task and a hash of the prompt and settings. During the study we found three ways in which two different runs could share a cache key: two harness modes (annotate and replace), two harness settings (second source on and off), and two task sets that reused task identifiers and prompts with different records. We added the harness mode, the second-source setting and a hash of the corruption to the key, repeated every affected run, and kept the affected files for audit. Every run also passed a check that its tool outputs contained the record

it was meant to receive; this check detected the third collision before any affected run entered an analysis. Runs failing the check are recorded as undelivered and excluded.

Deployment experiment

We ran Biomni (version 0.0.8) with DeepSeek V4.1 Flash (Together AI) and Gemini 3.8 Flash (Vertex AI, global endpoint). Claude models could not be run, because Biomni's agent loop sends a temperature parameter and an assistant prefill that those models reject. An intercepting proxy (mitmproxy 12.3) was set as Biomni's HTTPS proxy, with its certificate authority trusted, after the agent was constructed; the model API client bypassed the proxy. In a pilot of 25 tasks, every data request went to rest.uniprot.org or pubchem.ncbi.nlm.nih.gov and passed through the proxy (48 of 48). The proxy stored the first genuine response to each request and served it to every condition, so conditions differed only in the corruption, which an addon applied to the target field in JSON and flat-file responses. Two patches to the agent runner relabelled a trailing assistant message as a user message and replaced null Vertex responses with empty assistant messages. Each of 300 tasks ran once per condition and model. The first full run, in October 2026, ran under the author's own account with a data-lake path that contained the repository's name and that Biomni's system prompt shows to the model. An audit of its transcripts found that 44 Flash runs inferred a plausibility benchmark from the path, 98 Flash runs probed the local filesystem or fetched files from public repositories, and 2 used vocabulary from the study's task metadata; DeepSeek runs showed none of the first two and one of the third (Supplementary Note 1). The corrupted conditions were therefore rerun on 3 November 2026 under a separate macOS account with no access to the author's files, a neutral path, and a state file carrying only the task identifier and a slot number whose mapping to conditions was committed beforehand (commit 0019d72); the clean runs were retained from the first run. In the rerun no transcript named the study, its repository or its paper; 16 Flash runs inferred from the task itself that they were being tested, and the 17 Flash runs that probed the environment found nothing. We report the rerun throughout (600 corrupted runs per model). Excluding the 11 delivered plausible-value Flash runs that inferred they were being tested left plausible relay at 22% (57 of 265). Whether an agent computed length or mass from the sequence was scored from its executed code with a fixed pattern (a length taken of a sequence, Biopython's ProteinAnalysis or a molecular-weight function); this is an association, since agents chose whether to compute. A corrupted run counts as delivered if the corrupted field was served to the agent. Runs without a parseable answer count as neither propagated nor correct, as registered. For the third hypothesis, among the few relayed impossible values (4 for each model), Flash's were flagged in 2 of 4 against 89% in the benchmark (one-sided Fisher $p = 0.07$) and DeepSeek's in 3 of 4 against 98% ($p = 0.12$); for plausible values the one-sided p was 0.016 for DeepSeek and 0.99 for Flash, whose relayed values were flagged more often than in the benchmark (two-sided $p = 0.04$). For the derivation-route comparison, protein length and mass and compound molecular weight and heavy-atom count counted, as registered, as fields with an exact route, and all other fields formed the comparison group; odds ratios and 95% intervals are from the registered test, a logistic regression of relay on route class with task-clustered standard errors (DeepSeek 17.2, 95% interval 6.0–49.7; Flash 6.9, 3.3–14.3). Counting hydrogen-bond donors as exact-route fields gives 10.9 (4.2–28.4) for Flash; for DeepSeek every relayed other-field value was a donor count, leaving no other-field relay (0 of 123); with the Haldane–Anscombe correction the odds ratio is 106 (95% interval 6.5–1,740; Fisher exact $p < 10^{-10}$). The hypotheses, primary outcome, tests and exclusions were written as an OSF registration draft on 25 October 2026, after the pilots and before the first full run began (commit a2dc4a4 cites the draft); the draft was submitted unchanged on 1 November 2026 (OSF jmwab).

Override boundaries

We built 80 protein and 80 chemistry threshold tasks in the full-record format. In each corrupted run the target field and its paired field were scaled by the same factor (×0.9, 0.8, 0.7, 0.5, 0.3, 0.1, 0.02 or 0.005 when shrinking; ×1.1, 1.25, 1.5, 2, 3, 10, 50 or 200 when inflating), so that only the sequence or formula contradicted the value; 40 tasks per factor and direction. Each threshold lies midway between the true value and the value at the mildest factor. The factor at which half of runs overrode the corruption was estimated per model, domain and direction by logistic regression of override on the absolute log factor, with a weak ridge penalty on the slope, and a 95% interval from 400 task-cluster bootstrap resamples. Runs that failed at the API were rerun; runs without a parseable answer (97 of 6400) were excluded. On the 74 corrupted runs that succeeded in both the original sweep and the rerun, outcomes agreed in 64. The predictions registered for the new-field experiments were derived from the curves as first computed, in which 14 unparsed chemistry runs, at most 5 of 40 at any factor, were counted as overrides; we report those predictions as registered. Models: Opus 5, Sonnet 5, Gemini 4 Argon, Gemini 3.8 Flash and DeepSeek V4.1 Flash.

Coherent corruptions

For proteins we edited the sequence and recomputed every derived field: truncation to the scaled length (plausible, shrink), the first three residues (impossible), or extension to the scaled length with a segment of another benchmark protein that shares no 12-residue substring with the target (plausible, lengthen). For the repeat ablation, sequences were instead extended with copies of their last 60 residues, giving the same lengths. With repeats, Opus 5 and Argon relayed the wrong length in 40% and 44% of runs, against 76% and 70% with a natural segment; DeepSeek noted the repeat in 48 of 50 runs and relayed the wrong length in 92%. Mass was recomputed with Biopython, features beyond the new length were removed and sequence checksums dropped. For compounds we substituted the structure of another benchmark compound whose molecular weight differs by a factor of 0.72–0.83 or 1.17–1.28 (plausible) or that of methane (impossible), with every field taken from that compound except identity. Every coherent record passed our derivation checks. Arms: full record; summary record; summary record with the harness and no second source; summary record with the harness and the second source. Models: Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash. Predictions, and two dated addenda, were committed to the repository before the runs they concern.

Harness and second sources

The harness checks each record before the agent sees it. Protein length and mass are derived from the sequence (Biopython [39]; mass tolerance ±3 Da, above the largest deviation, 2.38 Da, among 360 calibration proteins), compound molecular weight from the structure (RDKit [40]; ±0.5 Da, above the largest deviation, 0.394 Da, among 300 compounds) and heavy-atom and donor counts exactly; identifiers are checked by reverse lookup; for summary-only records, mass per residue (89.6–130.3 Da) and mass per heavy atom (11.35–225.93 Da) must fall within bands calibrated on clean records. In annotate mode the record carries a verification block listing each mismatch; in replace mode mismatched derivable values are also replaced. The second sources were AlphaFold DB (sequence for a UniProt accession; 299 of 300 proteins covered, all 299 sequences identical to UniProt's, as expected because AlphaFold DB takes its sequences from UniProt; it is a current copy rather than an independent source) and NCI CACTUS (structure from a compound name; 283 of 300 resolved, formula identical to PubChem's for 270, which alone were used). Second-source checks used exact length, ±3 Da mass, ±0.5 Da molecular weight and exact heavy-atom

count, and raised no false alarm in 1,200 checks of clean fields. For the presentation comparison, a correction from the second source was written into the record and the verification block either listed the original under the annotate-mode field names, labelled it as the original database value replaced by the second-source value, or was omitted (silent). In the benchmark, replacing rather than annotating made no difference for DeepSeek (chemistry 21–22% against 22–24%; proteins at most 2%). On coherent errors, a harness message that all fields were consistent left protein relay at ceiling and lowered chemistry relay for Opus 5 (impossible 86% to 41%, plausible 100% to 77%).

New chemistry fields

Monoisotopic mass, topological polar surface area (TPSA) and PubChem complexity were taken from PubChem and classified by agreement with RDKit derivations from each record's structure: monoisotopic mass agreed within 0.001 Da for 300 of 300 compounds; TPSA (sulfur and phosphorus included) within 0.1 Å^2 for 80% (Spearman 0.996); complexity matched Bertz complexity for 1.7% (Spearman 0.936). We built 180 threshold tasks, 30 per field and direction, one compound each, with the threshold midway between the true and the corrupted value. Plausible values were scaled by 0.75–0.8 or 1.2–1.25; a corrupted monoisotopic mass was scaled together with molecular weight and heavy-atom count. Impossible values were 0 or 9×10^9 Da, -50 or 50,000 Å^2, and -100 or 10^7. Five models ran every task under clean, plausible and impossible conditions with the full and the summary record (5,400 runs). Predictions for monoisotopic mass were derived by code from each model's chemistry boundary curve, by log-linear interpolation, with prediction intervals combining the binomial error of both experiments. The follow-up varied agreement on the same 60 compounds: monoisotopic mass corrupted alone; molecular weight asked, with molecular weight and heavy-atom count scaled; molecular weight asked, with all three scaled. Predictions were fixed in commit 2059c3a and the follow-up's in commit 217ea9d; a 36-run cost-only pilot, disclosed in the registration, preceded the first and is excluded from all analyses.

Superseded UniProt sequences

For the 300 benchmark proteins we retrieved every sequence version from UniSave. 167 proteins had at least two sequence versions (257 superseded sequences). Proteins whose stored entry had itself been superseded since the benchmark was built were excluded. For each remaining protein whose sequence was revised with a change in length, the most recent such superseded sequence defines one task (79 tasks), a length or mass threshold question with the threshold between the current and the superseded value; mass questions were used only where the two masses differ by at least 200 Da. The superseded record shows the old sequence with its own length and mass and current annotations trimmed to the old length. Arms: full record; summary record; summary record with the harness and the second source. For cost, Opus 5 ran the full-record clean condition on the first 20 tasks. Across the three repeat samples, 90–100% of proteins gave the same outcome every time. The replace and presentation arms and the repeat samples were registered as an addendum before they ran (commits 6830c57 and 76e56b6). One-sided Fisher p for the direction comparison was 0.007, 0.044 and 0.116 in the three samples; in the first sample DeepSeek flagged the discrepancy in 24 of its 25 relayed runs.

Prompt variants

Four system prompts were compared on three cells, with predictions and tests fixed in commit 499d317 before the runs: (a) the protein benchmark tasks with the full record and a plausible paired-field corruption, so that the sequence contradicts the value (the 50 tasks whose target field

is length or mass); (b) arm C of the agreement follow-up, in which molecular weight is asked and three fields agree with the corrupted weight (60 compounds, three samples); and (c) coherent plausible protein errors with the full record, which nothing in the record contradicts (50 tasks). Models: Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash, with Gemini 3.8 Flash added on (b). The default prompt is given above. The no-tool-instruction variant omits the sentence "Use the provided tools to answer." The record-errors variant appends: "This database is known to contain errors. Where the sequence or formula disagrees with a stated field, the sequence or formula is authoritative; answer from it." The two-fields variant asks for a JSON line with three fields, "where answer is your best estimate of the true value, database_value is the value as stated in the record, and concerns lists anything about the data that looked wrong or surprising"; the answer field is graded as usual. Registered predictions: PA1, that the record-errors variant lowers relay on (a) by at least 20 points for every model, held for DeepSeek (100% to 55%) and not for Opus 5 (18% to 0%) or Argon (14% to 2%), whose default relay on (a) left fewer than 20 points to lose. PA2, that removing the tool instruction changes relay on (a) and (b) by fewer than 10 points for every model, failed: on (b) relay fell by 44, 29 and 57 points for Opus 5, Argon and Flash, and on (a) by 10, 2 and 14 points. PA3, that the two-fields variant lowers relay on (a) by at least 20 points and that the stated database value matches the record in at least 90% of runs, held in its first part only for DeepSeek (28 points) and failed in its second (the stated value matched the record in 0–28% of (a) runs and 63–100% of (b) runs). PA4, that no variant lowers relay on (c) below 80% for any model, failed for the two-fields variant (52% and 0% for Opus 5 and Argon) and held for the instruction variants. Tests were two-sided McNemar tests on task-paired outcomes, as registered; runs without a parseable answer were excluded (at most 12 of 50 in any cell).

Repetition versus consistency

The 60 compounds of the agreement follow-up were run with the full record, molecular weight asked, a plausible corruption of the weight and three samples per run, in four arms registered in commit 499d317: R0, the weight corrupted alone; R1, a second field (molecular_weight_reported) added to the record and scaled with the weight, so that a verbatim repeat agrees; R2, the same field left at its correct value, so that the repeat disagrees; and B, the follow-up's existing arm in which the heavy-atom count is scaled with the weight, so that a derived field agrees. Models: Opus 5, Gemini 4 Argon, Gemini 3.8 Flash and DeepSeek V4.1 Flash. RC1, that R1 exceeds R0, failed, with relay lower under R1 in every model (R0 57%, 65%, 86% and 64%; R1 37%, 37%, 76% and 59%; two-sided McNemar $p = 2 \times 10^{-5}$, 8×10^{-8}, 0.01 and 0.3; pooled 68% to 52%, 198 against 84 discordant pairs, $p < 10^{-10}$); the registered one-sided test in the predicted direction gives $p \approx 1$, and the two-sided values come from the same discordant counts. RC2, that B exceeds R1 by at least 10 points pooled over models, held (73% against 52%; one-sided $p < 10^{-10}$ pooled and at most 0.008 per model), but because R1 fell: B exceeded R0 by 2–7 points per model. RC3, that R2 falls below R0, held (0–3% against 57–86%; pooled $p < 10^{-10}$). The registration stated in advance that an unmet RC1 would mean that only derived agreement counts, and we adopt that reading. The added field caused no false overrides: on uncorrupted records every model answered correctly in 60 of 60 R1 runs.

Withholding harness

A third harness mode, registered in commit 499d317, removes from the record any derivable field that fails its derivation check, with no note, leaving the sequence or formula from which the agent can recompute it; identifiers, which are not derivable, are left untouched. Cells: the benchmark

protein and chemistry tasks with the full record, impossible and plausible, 100 tasks each, one sample; models Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash. BH1, that relay of numeric fields is at most 5% for every model, held: 0 of 709 numeric fields were relayed (proteins 0 of 259, compounds 0 of 450). BH2, that identifier fields are unchanged from the annotate arm, held for proteins (gene symbols and organisms relayed in 0–8% of runs, against 0–1%) and not for compound names, which were relayed less often (4–48%, against 40–92% in the annotate arm); the arms differ in record format (full against summary), which gives the agent the IUPAC name and structure to check a name against, so the comparison does not isolate the harness mode (Supplementary Table 5). Protein runs without a parseable answer (9, 3 and 29 of 200 for the three models, mostly empty final messages) were excluded and could not relay; DeepSeek answered 8 of its 171 parsed protein runs with a value that was neither the true nor the corrupted one. The registration stated in advance that this mode cannot reach coherent or historical errors, which pass derivation checks.

Memory probe

Each of Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash was asked, without tools, for the length in amino acids of each of the 79 historical proteins, given the accession, protein name and gene symbol, one call per protein (registered in commit 74415d3). The answer was classified as superseded (within one residue of the superseded length), current (within one of today's length), either (both, where the two differ by at most two residues), other or none, and joined by protein to the existing runs: the full and summary records without a second source, and the annotate arm with the second source, three samples. Memory held the superseded length for 7, 6 and 6 of the 79 proteins and the current length for 48, 53 and 50; 11–12 proteins per model were classed as either and excluded from the contrasts, as registered. P1, that memory matches the superseded length for at least 20% of proteins in at least one model, failed. P2, that with the second source relay is higher for superseded-memory than for current-memory proteins (one-sided Fisher, pooled over three samples), held for Opus 5 (13 of 21 against 2 of 144, $p < 10^{-10}$) and DeepSeek (17 of 18 against 38 of 150, $p = 1 \times 10^{-8}$), pooled over models $p < 10^{-10}$, and not for Argon (0 of 17 against 4 of 159). P3, that without a second source relay is at least 90% in both memory states for every model, held: current-memory proteins were relayed in 47 of 48, 48 of 53 and 50 of 50 full-record runs and in every summary-record run, and superseded-memory proteins in every run.

Identity-stripped and minimal records

Registered in commit 9234013, with an addendum (c4f1f43) before the runs it concerns. The anonymous record format replaces every accession in the question with an opaque handle (ENTRY-1, ENTRY-2), removes the gene-symbol lookup tool, and returns for each protein only its organism, length, mass and sequence, reduced from the full record after any corruption has been applied; the minimal format also keeps the accession, protein name and gene symbol but drops all annotation. Cells: the 79 superseded records without a second source (anonymous, one sample) and the 25 length-threshold benchmark tasks under clean, impossible and plausible conditions (anonymous, three samples; minimal and full, one sample). Models: Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash. IS1, that superseded values would be relayed in at least 90% of anonymous runs, held (97–100%). IS2, that plausible relay without identity would lie within 15 points of the full record, held for Argon (1% against 0%) and DeepSeek (93% against 100%) and failed for Opus 5 (68% against 4% on 25 tasks, first sample). IS3 (impossible relay at most 20%) and IS4 (clean accuracy at least 90%) held for all three models. IS5, registered after IS2 failed, that Opus 5's relay with the minimal record would lie within 15 points of the full or of the anonymous record, failed both

ways (28% against 4% and 68%; paired McNemar $p = 0.03$ and 0.02): identity and annotation each contributed (Supplementary Table 14).

Reasoning

Only the Anthropic provider exposes a reasoning setting, so the registered ablation (9234013) used Anthropic models: Opus 5 with thinking disabled on the full-record benchmark (first 50 tasks, impossible and plausible), against the adaptive-thinking runs, and Sonnet 5 with thinking disabled on the summary-record benchmark (200 protein and 100 chemistry tasks). TH1, that relay of contradicted values would be at least 10 points higher without thinking, held for Opus 5 on plausible values (8% to 46%; one-sided McNemar $p = 2 \times 10^{-6}$) and not on impossible values, which stayed near zero (0% to 2%), nor for Sonnet 5, whose relay of impossible values fell without thinking (76% to 70% for proteins, 73% to 63% for chemistry). TH2, that relay of plausible summary-record values would change by fewer than 10 points, held (92% to 94%; 100% to 100%). Supplementary Tables 15 and 16 give the counts and the request settings recorded for every model.

Prompt without the concerns field

The variant no_concerns (9234013) read: "You are a bioinformatics assistant. Use the provided tools to answer. End your reply with exactly one line of JSON: {"answer": ...}." It ran on the first 100 protein and 100 chemistry tasks of the summary-record benchmark, impossible and plausible, with Opus 5, Argon and DeepSeek, compared task by task with the registered prompt. NC1, that relay would stay within 10 points of the registered prompt, failed in 7 of 12 cells: relay rose by 11–43 points in six and fell by 23 in one (Argon, chemistry, impossible). For NC2 we registered a keyword detector for doubt voiced in the reply text, to be checked by hand on a random sample of 50 runs. Against the author's blind labels it agreed in 37 of 50 ($\kappa = 0.39$), adding no flags but missing 13 of the 20 runs labelled as voicing doubt. We therefore froze a classifier prompt for Haiku 4.5, which is not among the models judged (commit f5cae56), validated it once on the same 50 labels ($\kappa = 0.84$; 46 of 50, four additional flags, no misses), and applied it to the last 700 characters of reply text before the JSON line, as the labeller saw them, for every no-concerns run and the matching registered-prompt runs. Without the field, doubt was voiced in 36–84% of impossible-value runs and 0–29% of plausible-value runs, against concerns listed in 91–100% and 43–97% under the registered prompt. NC2, that unprompted rates would be below half of the solicited rates, held in 6 of 12 cells, all plausible except one (Supplementary Table 17). The keyword rates are reported as a lower bound.

Familiar duplicate field

The repetition arms were rebuilt with the copy under the field name average_mass, by definition the molecular weight (9234013): F1, weight and copy scaled together; F2, weight scaled and copy left at its true value; 60 compounds, full record, one sample; Opus 5, Argon, Gemini 3.8 Flash and DeepSeek; F1 also run clean. FD1, that F1 relay would exceed R1 by at least 10 points pooled, was not met at the registered threshold (64% against 56% of 233 paired runs; one-sided McNemar $p = 0.015$; Opus 5 +13, Argon +15, Flash +2, DeepSeek +2 points). FD2, that F1 would not exceed R0 by 10 points, held (149 against 148 of 234). FD3, that F2 would fall below R0, held (21 against 149 of 236; $p < 10^{-10}$). Clean accuracy with the copy present was 60 of 60 for every model (Supplementary Table 18).

Revision audit

For each of the 79 superseded sequences we asked why UniProt revised it. The superseded sequence was compared with the entry's current isoforms and located in UniParc through the EBI Proteins API (UniProt's own UniParc search returned only current sequences when the audit ran), and its cross-references were compared with the EMBL protein identifiers listed under "sequence caution" in the current entry. Sixteen superseded sequences remain current isoforms of their entries; 13 came from submissions the current entry flags as erroneous (erroneous initiation in 11 entries, frameshift in 3 and erroneous gene-model prediction in 2; an entry may carry several), and one from a submission with a miscellaneous discrepancy; one matches a current RefSeq or Ensembl model; and 48 are carried by no active record and have no recorded reason. Forty-six of the 79 changes are at the N-terminus. We call the 79 records superseded and only the 13 documented errors; relay did not differ materially by class (Supplementary Table 19).

Cue classification and validation

Two annotators labelled 300 concerns (200 protein, 100 chemistry), sampled from the benchmark traces, blind to model and to any automatic label, using written guidelines; a concern may cite several cues. Disagreements were resolved jointly into consensus labels. One annotator was the author and the other an independent annotator (Acknowledgements). The classifier is one of the evaluated models and therefore also classified its own concerns, under a prompt frozen before validation. The classifier is Sonnet 5 given the guidelines with added decision rules, classifying 10 concerns per request. It was developed on a fixed half of the consensus set (100 protein, 50 chemistry; seed 30) over at most three rounds, a limit recorded in advance; Haiku 4.5 and Sonnet 5 were compared in the last round under a pre-stated rule. The prompt and model were then frozen and committed, and run once on the held-out half, with a pass criterion of Cohen's [41] $\kappa \geq 0.6$ for cross-field, recall and rule cues in both domains. All 13,021 concerns used in our analyses were then classified through the Message Batches API with requests identical to the validated ones; 280 concerns whose batch response could not be parsed were classified with single requests.

Registrations

Throughout, 'registered' means that predictions, outcomes and exclusions were fixed in a dated repository commit before the runs they concern: the new-field experiment (2059c3a), its agreement follow-up (217ea9d), the historical-error experiment (ac99248), the robustness addendum (6830c57, 76e56b6), the coherent corruptions (dated addenda), the memory probe (74415d3), the prompt ablation, repetition and withholding experiments (499d317) the deployment rerun's slot-to-condition schedule (0019d72), the identity, reasoning, prompt and duplicate-field experiments (9234013, addendum c4f1f43) and the unprompted-doubt classifier prompt (f5cae56). Commit and run timestamps are listed in Supplementary Note 2. The deployment hypotheses were drafted on OSF before the full runs and submitted unchanged on 1 November 2026 (osf.io/jmwab).

Data availability

All run traces, task files, corrupted records, proxy logs, cue labels and annotator labels are archived at Zenodo (10.5281/zenodo.23937493), which snapshots release v1.1-nmi of the repository below.

Code availability

Code, prompts, registrations and analysis scripts are available at `https://github.com/Quantum535/bioagent-plausibility` (release v1.1-nmi) and in the Zenodo archive above.

Prior presentation

A preliminary account of the benchmark results (Extended Data Fig. 1) was presented at the AgenticLS workshop at ICML 2026; the deployment, boundary, coherent-corruption, new-field, historical-error, harness, presentation, prompt, repetition, memory, withholding, identity, reasoning and duplicate-field experiments and the revision audit are new to this paper.

Use of AI tools

AI tools (Claude, Anthropic) were used to assist in writing analysis scripts, in analysis, and in drafting and editing text. The author reviewed and is responsible for all code, analyses, figures and text.

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Author contributions

G.X. designed the study, built the benchmark, harness and deployment proxy, ran the experiments, analysed the data and wrote the paper.

Competing interests

The author declares no competing interests.

References

[1] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

[2] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. doi: 10.1038/s42256-024-00832-8.

[3] Kexin Huang, Serena Zhang, Hanchen Wang, et al. Autonomous biomedical research with an artificial intelligence agent. *Science*, 2026. doi: 10.1126/science.adz4351. Article eadz4351.

[4] Z. Wang, Q. Jin, C.-H. Wei, et al. GeneAgent: self-verification language agent for gene-set analysis using domain databases. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02748-6.

[5] Alexandra M. Schnoes, Shoshana D. Brown, Igor Dodevski, and Patricia C. Babbitt. Annotation error in public databases: Misannotation of molecular function in enzyme superfamilies. *PLoS Computational Biology*, 5(12):e1000605, 2009. doi: 10.1371/journal.pcbi.1000605.

[6] Steven E. Brenner. Errors in genome annotation. *Trends in Genetics*, 15(4):132–133, 1999.

[7] Walter R. Gilks, Benjamin Audit, Daniela De Angelis, Sophia Tsoka, and Christos A. Ouzounis. Modeling the percolation of annotation errors in a database of protein sequences. *Bioinformatics*, 18(12):1641–1649, 2002. doi: 10.1093/bioinformatics/18.12.1641.

[8] Craig E. Jones, Alfred L. Brown, and Ute Baumann. Estimating the annotation error rate of curated GO database sequence annotations. *BMC Bioinformatics*, 8:170, 2007. doi: 10.1186/1471-2105-8-170.

[9] The UniProt Consortium. UniProt: the universal protein knowledgebase in 2025. *Nucleic Acids Research*, 53(D1):D609–D617, 2025. doi: 10.1093/nar/gkae1010.

[10] Rasko Leinonen, Francesco Nardone, Weimin Zhu, and Rolf Apweiler. UniSave: the UniProtKB sequence/annotation version database. *Bioinformatics*, 22(10):1284–1285, 2006. doi: 10.1093/bioinformatics/btl105.

[11] Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. Entity-based knowledge conflicts in question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7052–7063, 2021. doi: 10.18653/v1/2021.emnlp-main.565.

[12] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8541–8565, 2024. doi: 10.18653/v1/2024.emnlp-main.486.

[13] Kevin Wu, Eric Wu, and James Zou. ClashEval: Quantifying the tug-of-war between an LLM's internal prior and external evidence. In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, 2024.

[14] Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. Benchmarking large language models in retrieval-augmented generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17754–17762, 2024.

[15] Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *International Conference on Learning Representations*, 2024.

[16] Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. Context-faithful prompting for large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 14544–14556, 2023. doi: 10.18653/v1/2023.findings-emnlp.968.

[17] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, 2023.

[18] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638.

[19] Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *34th USENIX Security Symposium*, pages 3827–3844, 2025.

[20] Mrinank Sharma, Meg Tong, Tomasz Korbak, et al. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, 2024.

[21] Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, et al. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.

[22] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems 36*, 2023.

[23] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.

[24] Yujia Qin, Shihao Liang, Yining Ye, et al. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *International Conference on Learning Representations*, 2024.

[25] Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive APIs. In *Advances in Neural Information Processing Systems 37*, 2024.

[26] Xiao Liu, Hao Yu, Hanchen Zhang, et al. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations*, 2024.

[27] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. In *International Conference on Learning Representations*, 2025.

[28] Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

[29] Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, 2024.

[30] Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. Identifying the risks of LM agents with an LM-emulated sandbox. In *International Conference on Learning Representations*, 2024.

[31] Hoyeol Yang, Woojung Song, Taewon Kim, Jonghyun Song, Seoyeon Park, and Yohan Jo. Agents' overreliance on unreliable tools. *arXiv preprint arXiv:2609.05587*, 2026.

[32] Joe Needham, Giles Edkins, Govind Pimpale, Henning Bartsch, and Marius Hobbhahn. Large language models often know when they are being evaluated. *arXiv preprint arXiv:2505.23836*, 2025.

[33] Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A. Shoemaker, Paul A. Thiessen, Bo Yu, Leonid Zaslavsky, Jian Zhang, and Evan E. Bolton. PubChem 2025 update. *Nucleic Acids Research*, 53(D1):D1516–D1525, 2025. doi: 10.1093/nar/gkae1059.

[34] Xiang Zhang, Juntai Cao, and Chenyu You. Counting ability of large language models and impact of tokenization. *arXiv preprint arXiv:2410.19730*, 2024.

[35] Mihaly Varadi, Damian Bertoni, Paulyna Magana, et al. AlphaFold protein structure database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*, 52(D1):D368–D375, 2024. doi: 10.1093/nar/gkad1011.

[36] John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021. doi: 10.1038/s41586-021-03819-2.

[37] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[38] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008.

[39] Peter J. A. Cock, Tiago Antao, Jeffrey T. Chang, et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422–1423, 2009. doi: 10.1093/bioinformatics/btp163.

[40] Greg Landrum et al. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2025.

[41] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.



Figure 1: **Study design and a deployed agent. a**, Corrupted values were injected between the database and the agent, in a tool layer or, for the deployed agent, an intercepting proxy; the agent's answer and concerns were scored. Bottom: the forms a corrupted record took (illustrative values). **b**, Biomni with live UniProt and PubChem APIs: share of corrupted runs in which the value was relayed, caught, or never requested (relayed/delivered at right). **c**, Relay of plausible values by field (shaded rows, compounds; filled circles, fields counted as having an exact derivation route (Methods); open diamonds, other fields), with 95% Wilson intervals; numbers give relayed/delivered runs.

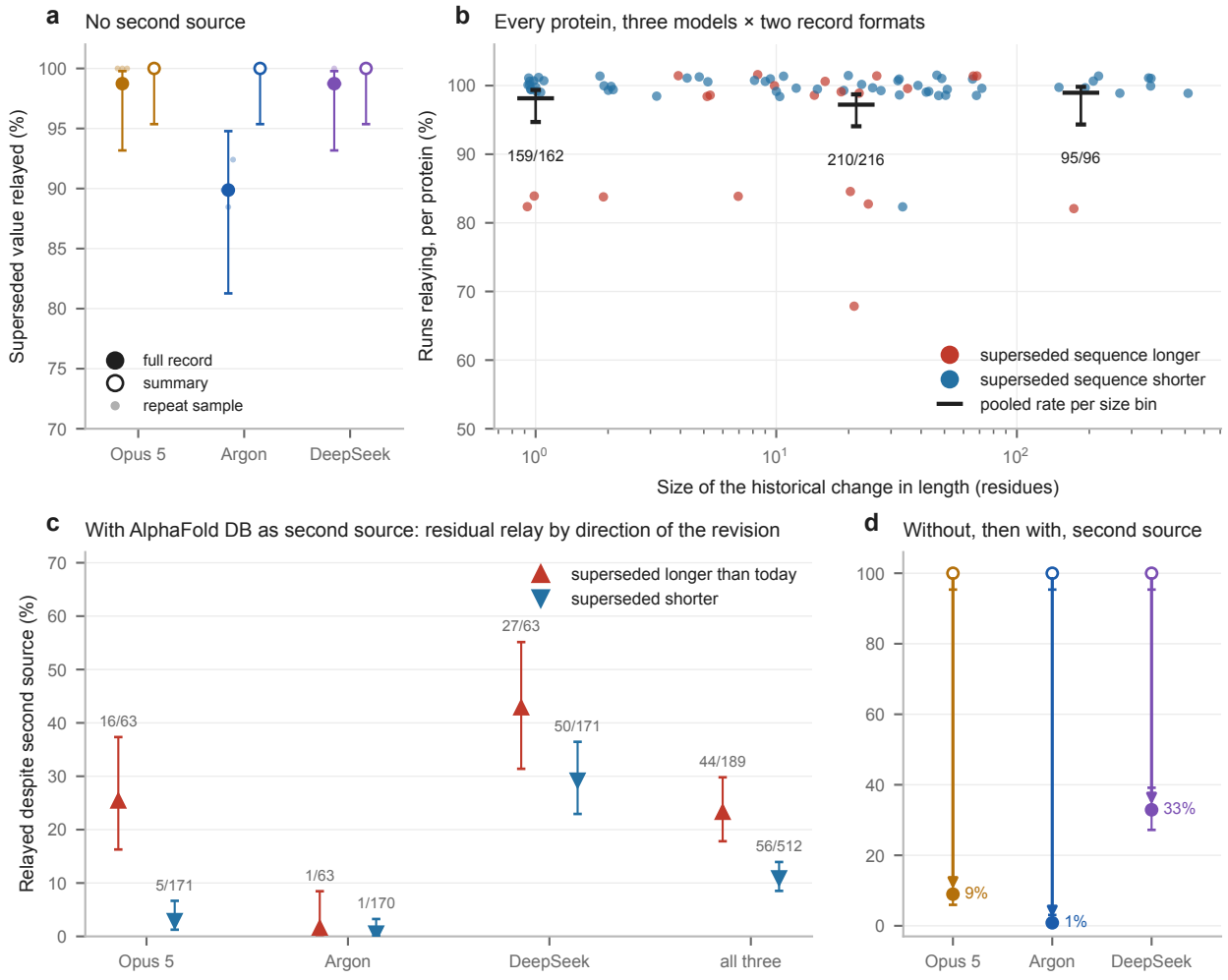


Figure 2: **Superseded UniProt sequences.** **a**, Relay of the superseded value with the full (filled) and summary (open) record; small dots, repeat samples with the full record. **b**, For each of the 79 proteins, the share of six model–format runs relaying the superseded value against the size of the historical change in length, coloured by whether the superseded sequence was longer or shorter than today's; black bars, pooled rate per size bin with 95% interval. **c**, With AlphaFold DB as second source, residual relay on covered proteins by direction of the revision (three samples pooled). **d**, Relay without (open) and with (filled) the second source.

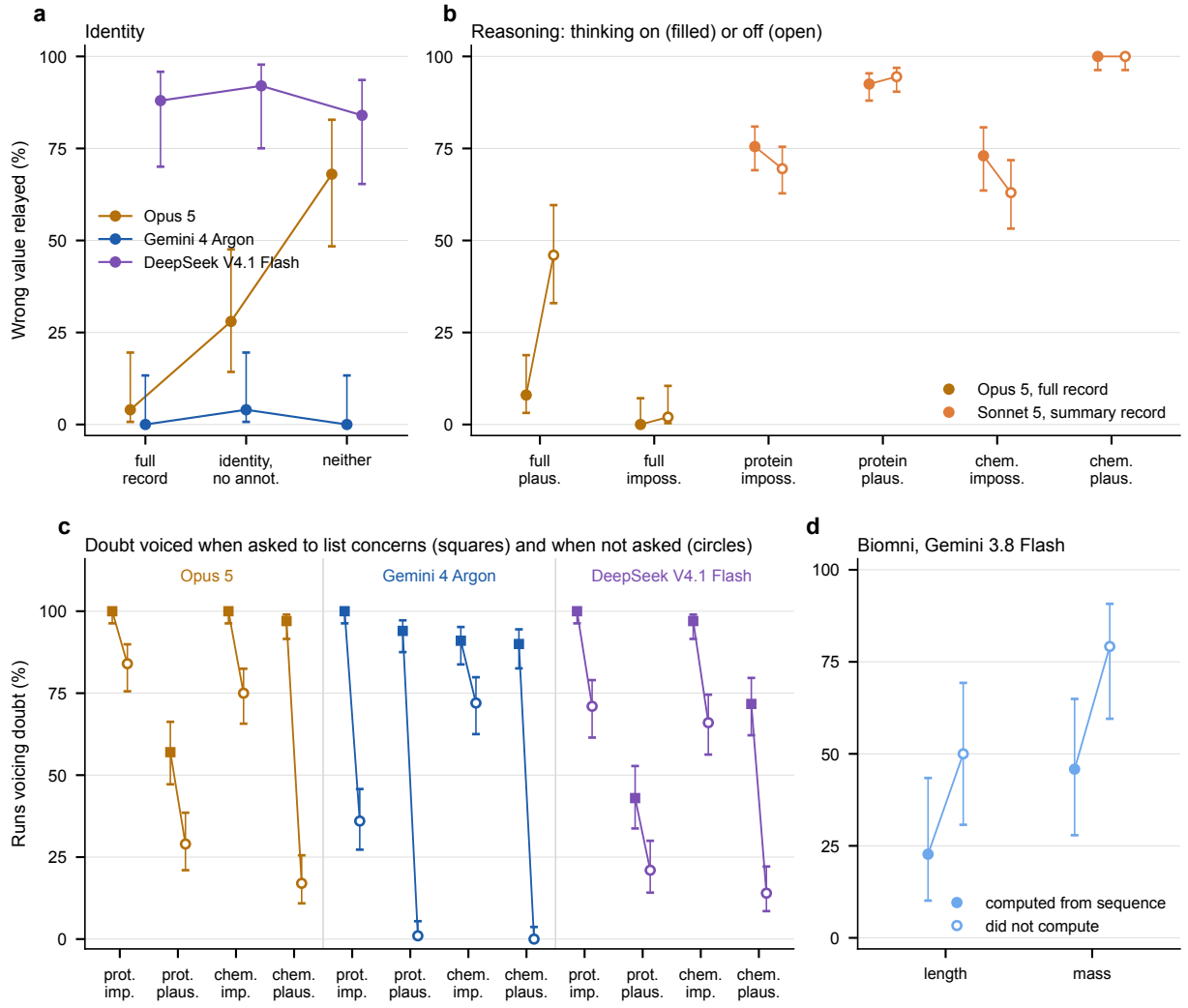


Figure 3: **What the agent brings to a contradiction.** **a**, Identity: relay of a plausible wrong length that the full record contradicts, with the full record, with identity but no annotation (accession, name, gene, organism, length, mass, sequence), and with neither; 25 tasks per model. **b**, Reasoning: relay with and without thinking for Opus 5 with the full record (first 50 benchmark tasks) and for Sonnet 5 on impossible and plausible values (summary record). **c**, Unprompted doubt: runs in which the agent voiced doubt about the data with and without the request to list concerns, by model, domain and condition, scored by a classifier validated against human labels. **d**, Biomni: relay of plausible length and mass errors by Gemini 3.8 Flash when it computed the value from the sequence and when it did not. Markers give rates with 95% Wilson intervals.

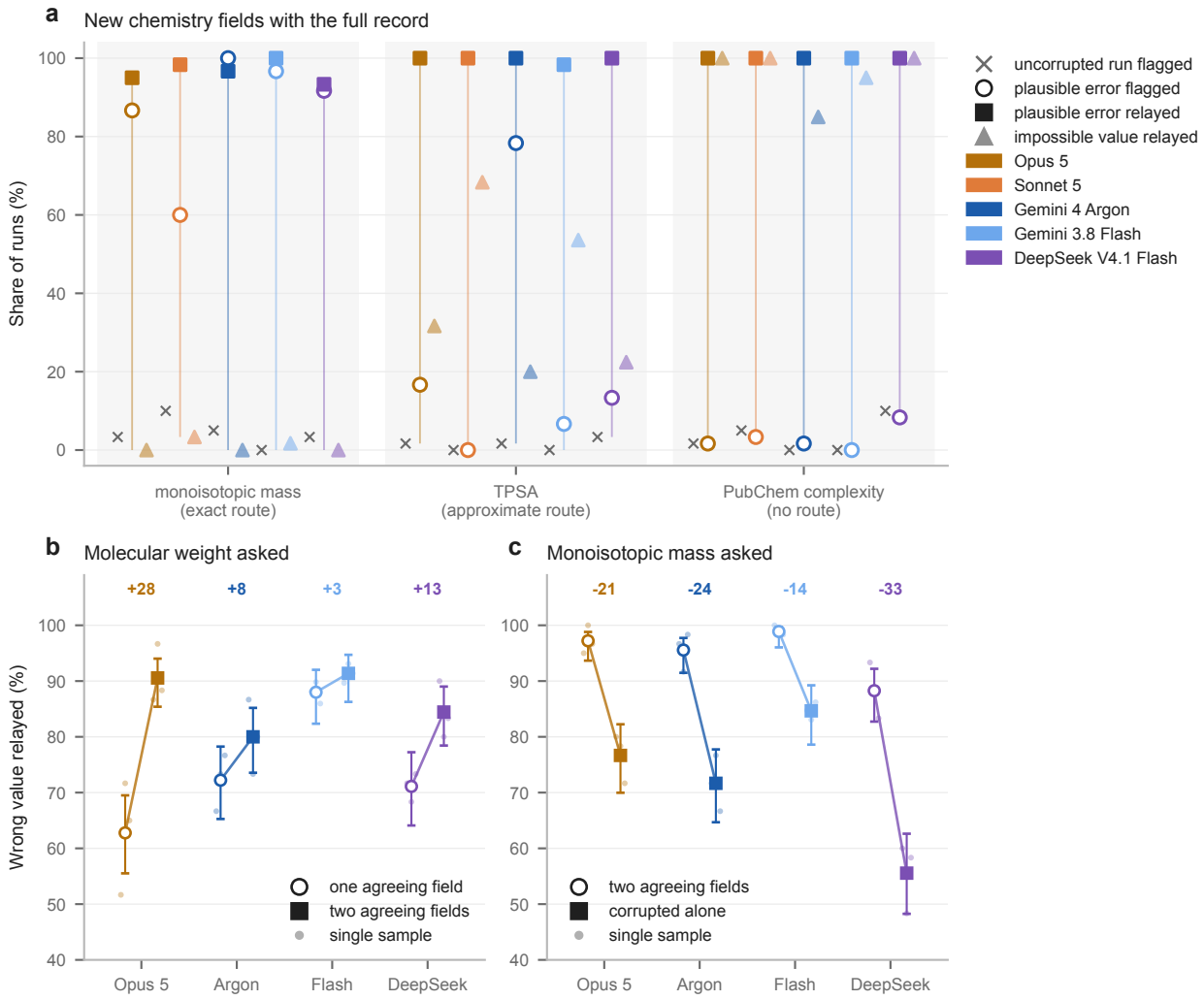


Figure 4: **A derivation route raises detection; agreement decides override. a**, New chemistry fields with the full record, per model: uncorrupted runs raising a concern (crosses); plausible-error runs raising a concern (open circles) and relaying the value (squares); impossible-value runs relaying it (triangles). **b**, Molecular weight asked: relay when the corrupted weight agreed with one or with two other fields. **c**, Monoisotopic mass asked: relay when the corrupted mass agreed with two other fields or was changed alone. Large markers pool three samples (95% Wilson intervals); small dots are single samples; numbers give the change in percentage points.

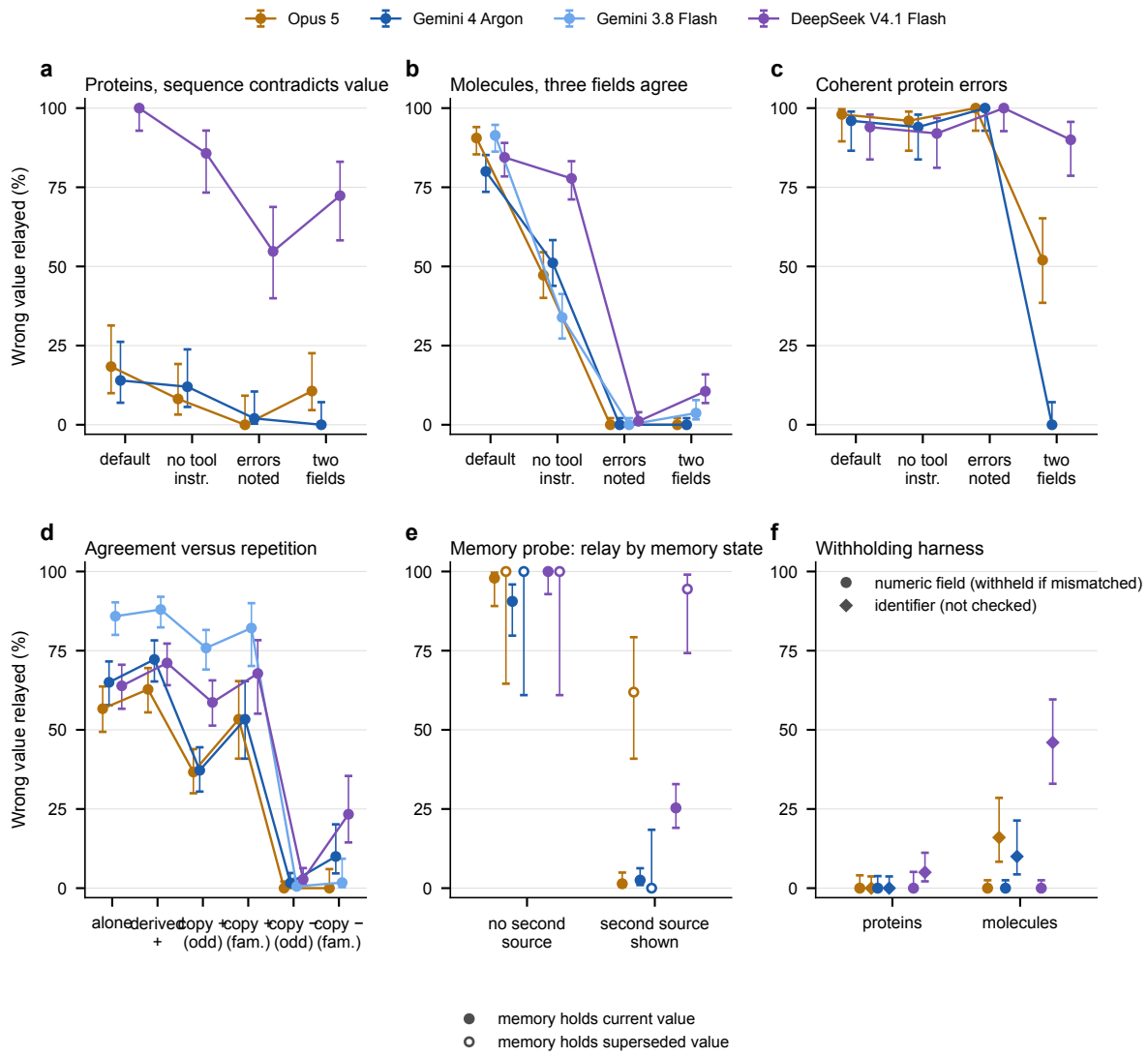


Figure 5: **What turns detection into override.** **a**–**c**, Relay under four system prompts: the registered default; without the instruction to use the tools (no tool instr.); with a note that the database contains errors and the sequence or formula is authoritative (errors noted); with the answer and the record's value asked for as separate fields (two fields) on three cells: protein tasks whose full record contradicts the value (**a**), the chemistry agreement cell with three fields agreeing with the wrong weight, three samples (**b**), and coherent protein errors that nothing in the record contradicts (**c**). **d**, Agreement versus repetition: relay of a wrong molecular weight alone (alone), with a derived field (heavy-atom count) that agrees (derived +), and with a verbatim copy that agrees (copy +) or disagrees (copy −) under an unfamiliar (molecular weight reported; odd) or familiar (average mass; fam.) field name; 60 compounds, three samples, familiar-name arms one sample. **e**, Memory probe: relay of the superseded value for proteins whose memory held the current (filled) or the superseded (open) length, without a second source (full record) and with the second source shown (three samples). **f**, Withholding harness: relay of corrupted numeric fields, withheld when they fail the derivation check (circles), and of identifiers, which the harness does not check (diamonds); impossible and plausible pooled. Markers give relay with 95% Wilson intervals; lines connect one model.

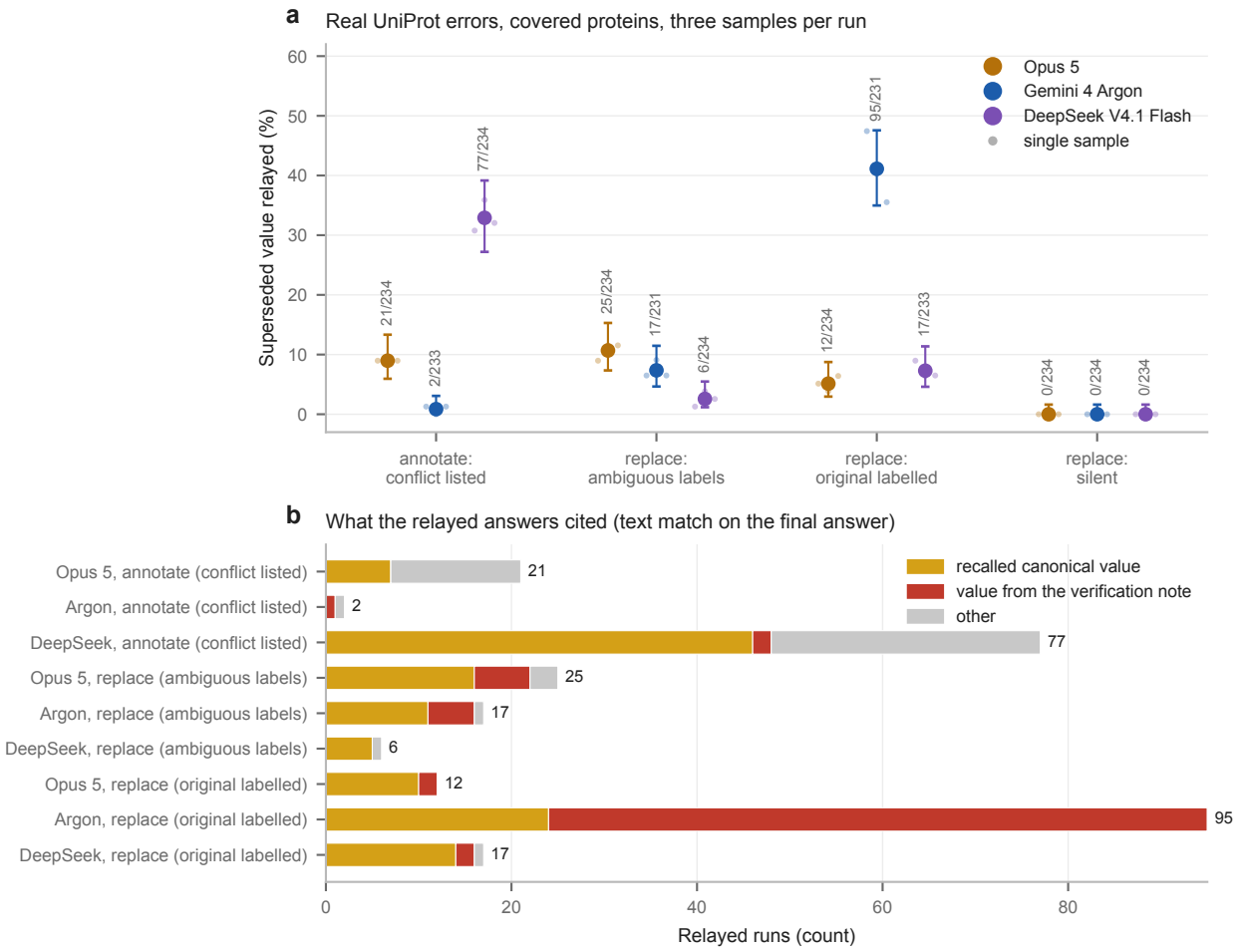
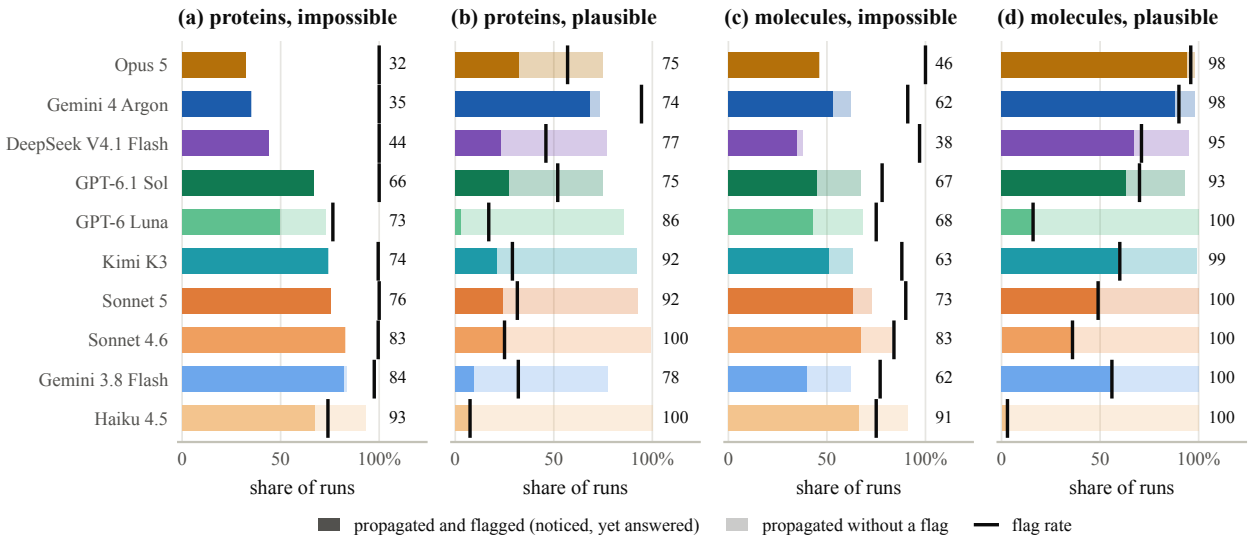
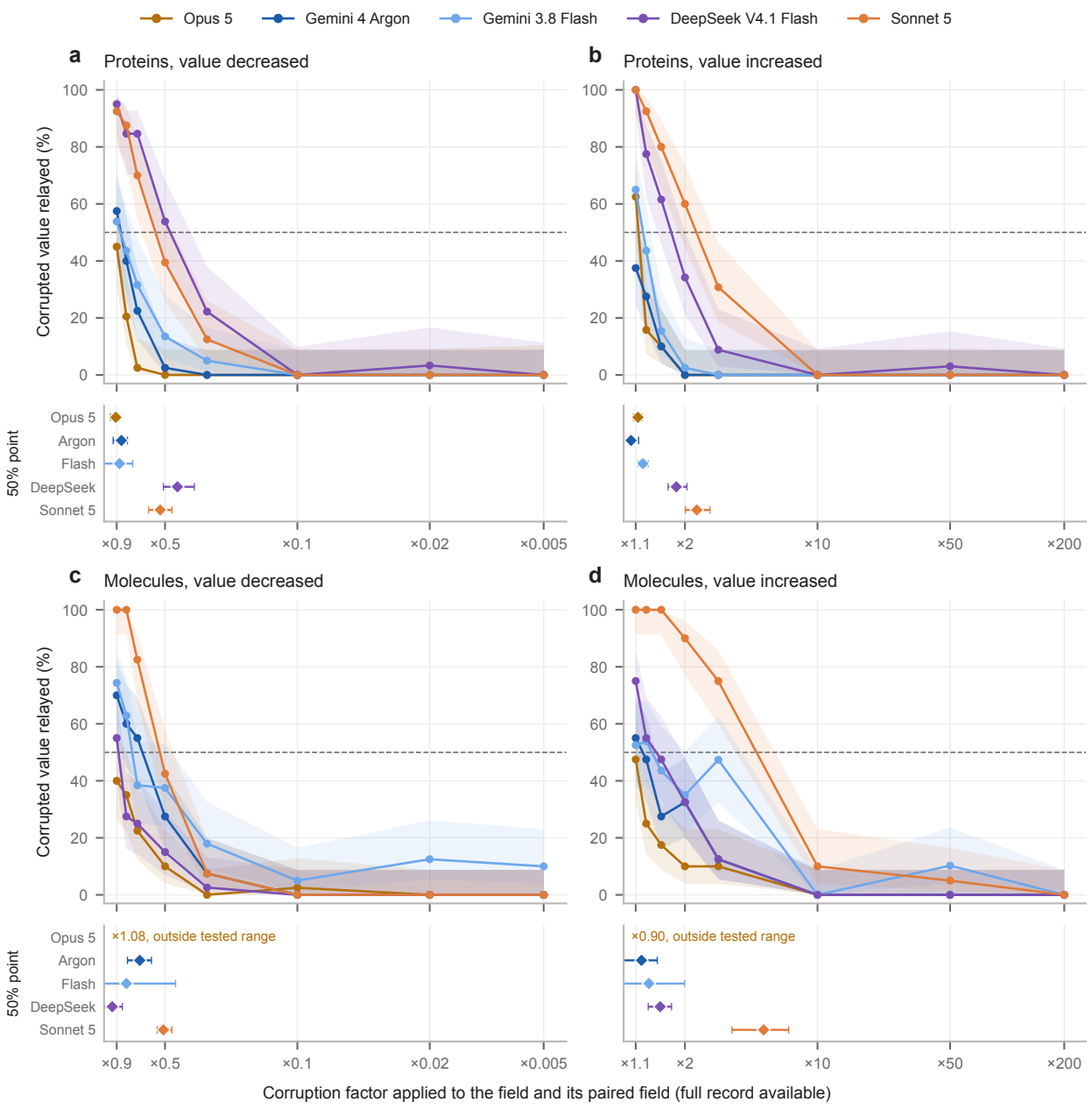


Figure 6: **Correcting the record works; showing the conflict does not. a**, Relay of superseded values on covered proteins under four presentations of the second source (three samples per run; 95% Wilson intervals; small dots, single samples; counts above). **b**, What the relayed answers cited, by a text match on the final answer: a recalled canonical value, the original or observed value shown in the verification note, or neither.



Extended Data Fig. 1: **Noticed yet answered.** Share of runs in which each model relayed the corrupted value into its final answer, for impossible and plausible values in the protein (**a**, **b**) and chemistry (**c**, **d**) benchmarks. Dark segments: relayed runs in which the agent also raised a concern; light segments: relayed without a concern; black ticks: share of all runs raising a concern. Runs used the registered prompt, which asks the agent to list concerns. Numbers give the relay rate (%).



Extended Data Fig. 2: **Override boundaries.** Share of runs relaying a corrupted value as the corruption grows, with the full record available and the paired field scaled alike, for proteins (**a**, value decreased; **b**, increased) and molecules (**c**, decreased; **d**, increased). Bands: 95% Wilson intervals (30–40 tasks per point after excluding runs without a parseable answer). Strips below each panel: fitted factor at which half of runs relay the value, with its 95% interval; estimates outside the tested range are given as numbers.